1. Problem Modeling

Urban rail transit system

Infrastructure data, passenger flow data, operational rules



Optimization model

Objectives

- Minimize traction energy consumption
- Minimize average passenger waiting time

Decision variables

- Train departure headways $h_{l,n}^{d}$
- Passenger flow allocation ratios $x_{ij,r}^{t}$

Constraints

- Departure interval constraints
- Last departure time constraints
- Passenger flow guidance constraints

2. Ordinal Optimization-Based Solution Method

Step 1 Extraction of the Representative Set

- Randomly sample N feasible solutions from the solution space based on constraints to form the representative set S.

Step 2 Rough Model Computation

- Evaluate each solution in S using a simplified (rough) model with fixed passenger flow and larger simulation time step size.

Step 3 Determine the VOPC Type

- Identify the vector ordered performance curve (VOPC) type (Flat, Neutral, or Steep) based on the distribution of solutions.

Step 4 Calculation of Noise Error

- Compute the normalized noise level between rough model and detailed model.

Step 5 Selection of the Candidate Set

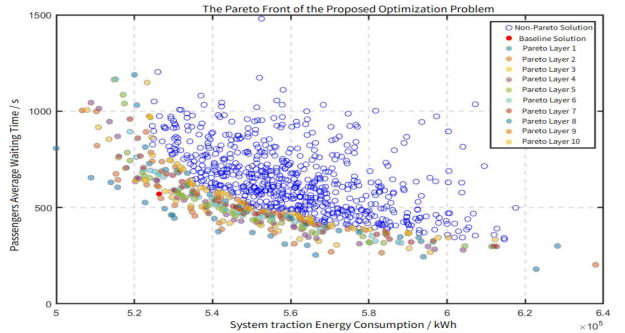
- Determine the number of Pareto layers to be observed (l) according to VOPC type, error level, and parameters m, k, α.
- Select the first l Pareto layers generated by the rough model as the candidate set.

Step 6 Detailed Model Simulation

- Perform detailed simulations on the candidate set, and select the best k solutions as the final plan.

3. Solution Evaluation and Case Study

Pareto front and solution quality assessment



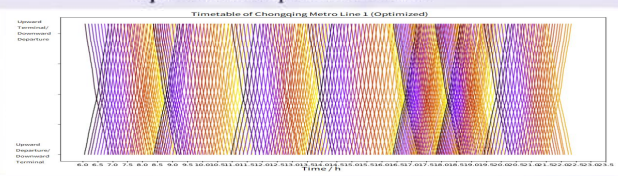
Case study: Chongqing Metro Lines 1, 2, and 3

- Network scale: 89 stations, 97 sections
- Representative set size: 1,000
- VOPC type: Steep type
- Noise error range: [−0.5, 0.5]
- By observing the first 2 Pareto layers in the rough model, 9 solutions from the first 3 Pareto layers in the detailed model can be selected with a confidence level of $\alpha = 95\%$.
- Comparison with baseline scheme:

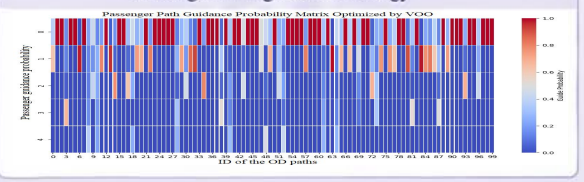
Plans	Full-day traction energy consumption/kWh				Passenger average waiting time/min
	Line 1	Line 2	Line 3	All	
Baseline	38581	37402	42255	157083	2.667
Our method	32029	27448	32271	123508	2.093

4. Optimization Results

Optimized train operation timetable



Passenger flow guidance strategy



Collaborative optimization results

- ✓ Optimized train scheduling scheme
- ✓ Optimized passenger flow guidance strategy
- ✓ Reduced system traction energy consumption
- ✓ Lower average passenger waiting time
- ✓ Assessable solution quality

Fig. 1. Overall framework of the ordinal optimization-based method for the collaborative optimization of train scheduling and passenger flow guidance in urban rail transit.